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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 7	
TITLE : Inheritance and Interfaces	

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Exercise

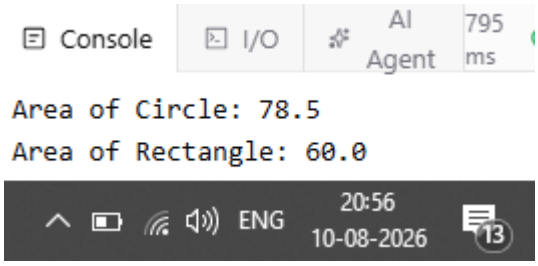
1.Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

Code:

```
class Shape {
double area() {
return 0;
}
}
class Circle extends Shape {
double radius;
Circle(double radius) {
this.radius = radius;
}
double area() {
return 3.14 * radius * radius;
}
}
class Rectangle extends Shape {
double length;
double width;
Rectangle(double length, double width) {
this.length = length;
this.width = width;
```

```
}
double area() {
return length * width;
}
}
public class Main {
public static void main(String[] args) {
Circle c = new Circle(5);
Rectangle r = new Rectangle(10, 6);
System.out.println("Area of Circle: " + c.area());
System.out.println("Area of Rectangle: " + r.area());
}
}
```

Output:



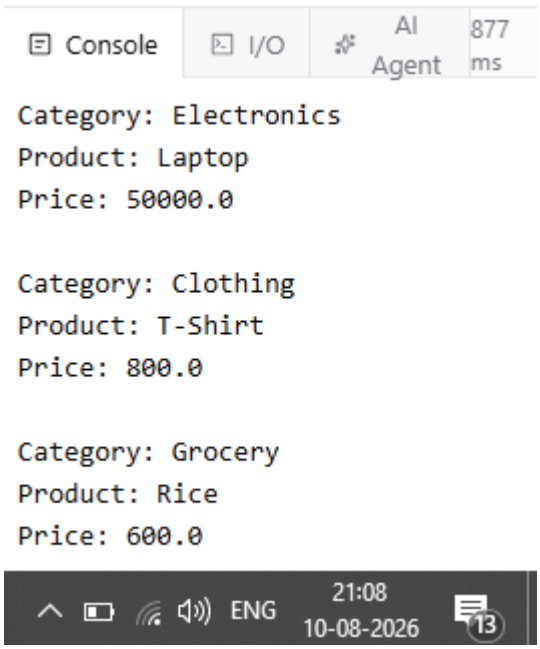
2.Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Code:

```
class Product {
String name;
double price;
Product(String name, double price) {
this.name = name;
this.price = price;
}
void display() {
System.out.println("Product: " + name);
System.out.println("Price: " + price);
}
}
interface ProductDetails {
void showDetails();
}
class Electronic extends Product implements ProductDetails {
```

```
Electronic(String name, double price) {
super(name, price);
}
public void showDetails() {
System.out.println("Category: Electronics");
display();
}
}
class Clothing extends Product implements ProductDetails {
Clothing(String name, double price) {
super(name, price);
}
public void showDetails() {
System.out.println("Category: Clothing");
display();
}
}
class Grocery extends Product implements ProductDetails {
Grocery(String name, double price) {
super(name, price);
}
public void showDetails() {
System.out.println("Category: Grocery");
display();
}
}
public class Main {
public static void main(String[] args) {
Electronic e = new Electronic("Laptop", 50000);
Clothing c = new Clothing("T-Shirt", 800);
Grocery g = new Grocery("Rice", 600);
e.showDetails();
System.out.println();
c.showDetails();
System.out.println();
g.showDetails();
}
}
```

Output:



Conclusion

Inheritance helps in reusing common properties and methods from an existing class. Interfaces provide abstraction by defining common methods that different classes can implement. Using both makes the program more organized and reduces code duplication. It also improves flexibility and makes the code easier to maintain. Thus, inheritance and interfaces are useful concepts for achieving code reusability and abstraction in Java.